

DevOps: Using Ansible to Deploy Cacti for Monitoring

In this third article in the DevOps series, we will install and set up Cacti, a free and open source Web-based network monitoring and graphing tool, using Ansible.



Cacti is written in PHP and uses the MySQL database as a backend. It uses the RRDtool (Round-Robin Database tool) to handle time series data and has built-in SNMP support. Cacti has been released under the GNU General Public License.

Setting up Cacti

We will use a CentOS 6.8 virtual machine (VM) running on KVM to set up Cacti. Just for this demonstration, we will disable SELinux. You will need to set the following in `/etc/selinux/config` and reboot the VM:

```
SELINUX=disabled
```

When used in production, it is essential that you enable SELinux. You should then test for Internet connectivity from within the VM.

The Ansible version used on the host Parabola GNU/Linux-libre x86_64 is 2.2.1.0. The `ansible/inventory/kvm/` directory structure is shown below:

```
ansible/inventory/kvm/inventory
ansible/inventory/kvm/group_vars/all/all.yml
```

The IP address of the guest CentOS 6.8 VM is provided in the inventory file as shown below:

```
centos ansible_host=192.168.122.98 ansible_connection=ssh
```

```
ansible_user=root ansible_password=password
```

Add an entry for 'centos' in the `/etc/hosts` file as indicated below:

```
192.168.122.98 centos
```

The contents of the `all.yml` for use with the playbook are as follows:

```
---
mysql_cacti_password_hash: "{{ vault_mysql_cacti_password_hash }}"
```

```
mysql_username: "{{ vault_mysql_user }}"
mysql_password: "{{ vault_mysql_password }}"
```

The `Cacti.yml` playbook is located in the `ansible/playbooks/configuration` folder.

Vault

Ansible provides the Vault feature, which allows you to store sensitive information like passwords in encrypted files. You can set the `EDITOR` environment variable to the text editor of your choice, as shown below:

```
$ export EDITOR=nano
```

In order to store our MySQL database credentials, we will